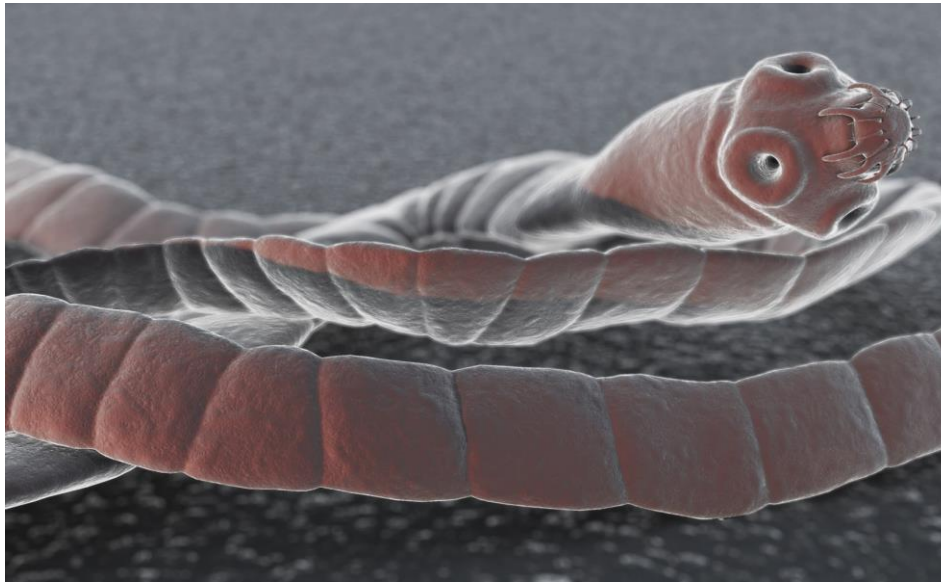


General properties of Cestodes and nematodes



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First stage

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CLASSIFICATION

HELMINTHES



General Characteristics of Cestodes

- Cestodes (Greek keston—girdle or ribbon) are multisegmented, dorsoventrally flattened tape-like worms.
- Their sizes vary from a few millimeters to several meters.
- The adult worms are found in the small intestine of humans (*Taenia solium*, *Taenia saginata*) or in the small intestine of dogs (*Echinococcus granulosus*). They remain attached to the intestinal mucosa with their suckers.
- The head carries suckers and some also have hooks.
- tapeworms lack **digestive system**.

- Cestodes are **hermaphrodites** (monoecious) and every mature segment contains both male and female sex organs.
- Humans usually acquire the infection when undercooked meat or fish containing **the larvae** is ingested.
- In two important human diseases, cysticercosis and hydatid disease, it is **the eggs** that are ingested and the resulting larvae cause the disease.
- Excretory and nervous systems are present.
- They are oviparous.

Classification of Cestodes

- Cestodes belong to Phylum Platyhelminthes and class Cestoidea.
- The class Cestoidea includes 2 orders:
 - 1- Pseudophyllidea (scolex has grooves) called bothria.
 - 2- Cyclophyllidea a (scolex has suckers)



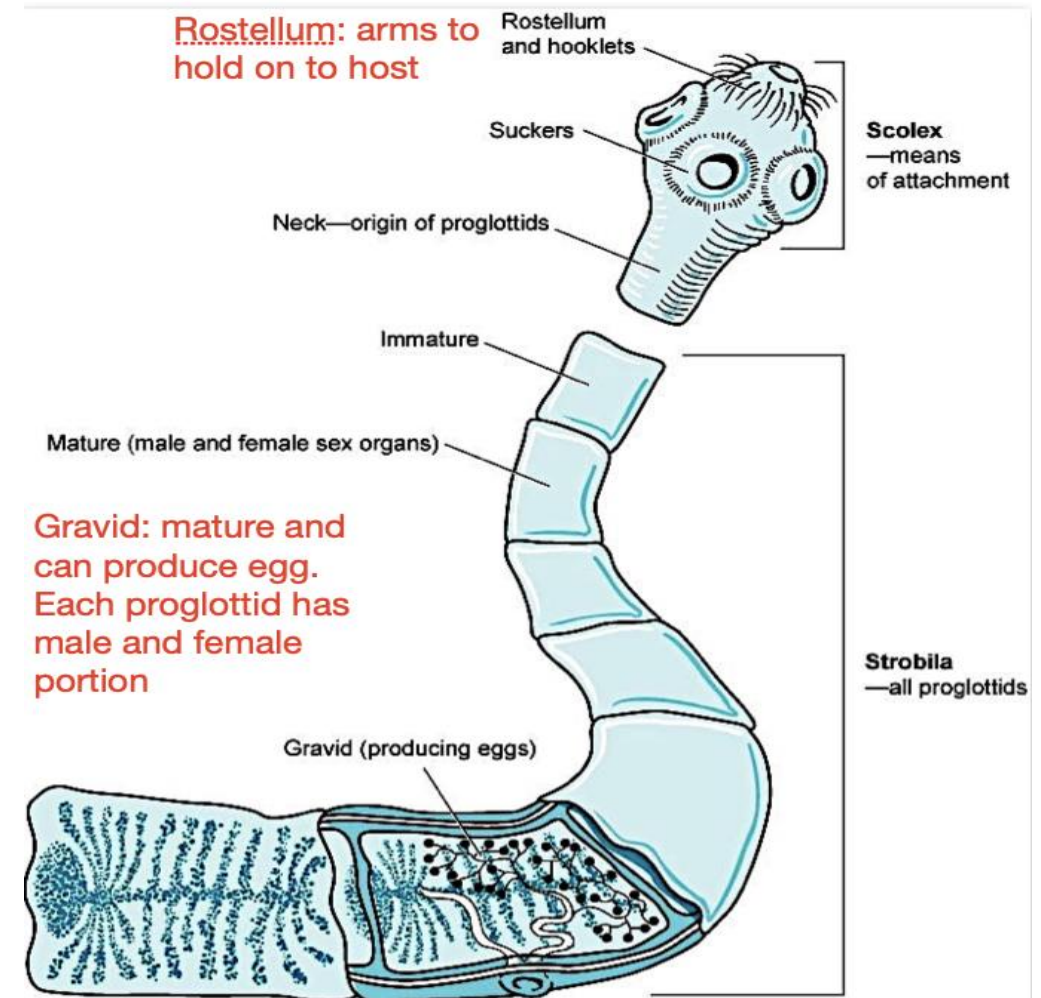
Cyclophyllidea - *Taenia*
sp.



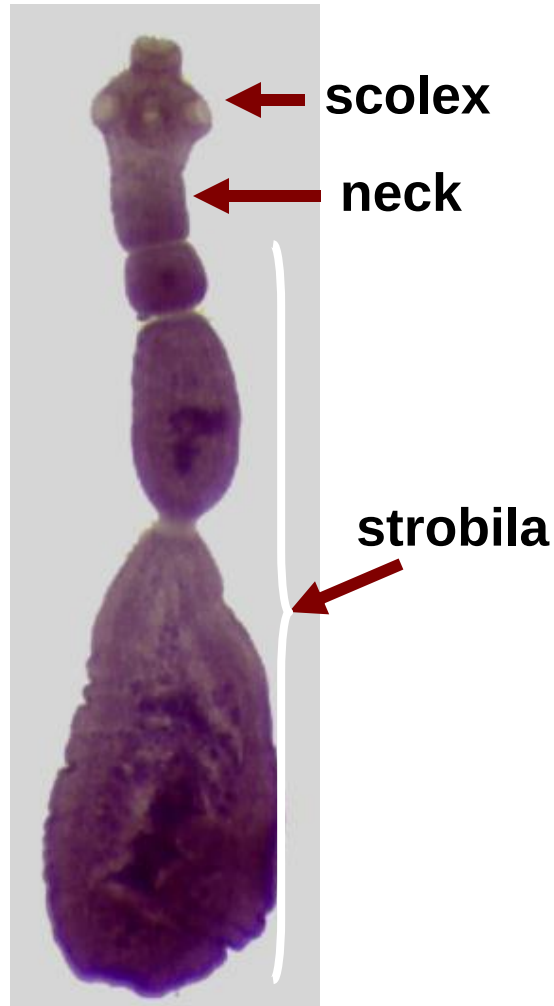
Pseudophyllidea --
Diphyllobothrium
sp.

Tapeworms (adult Worms)

- The adult worm consists of 3 parts:
- Head (scolex)
- Neck
- Trunk (strobila)

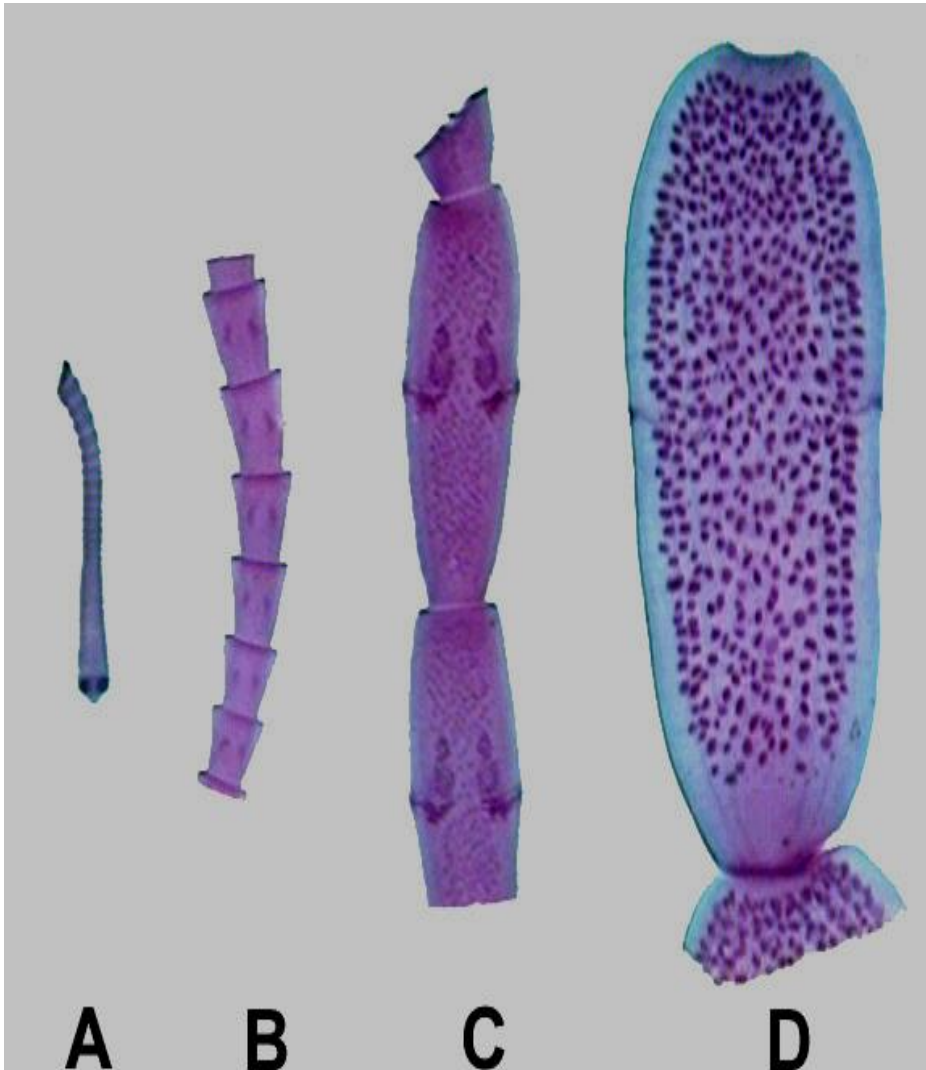


Tapeworms



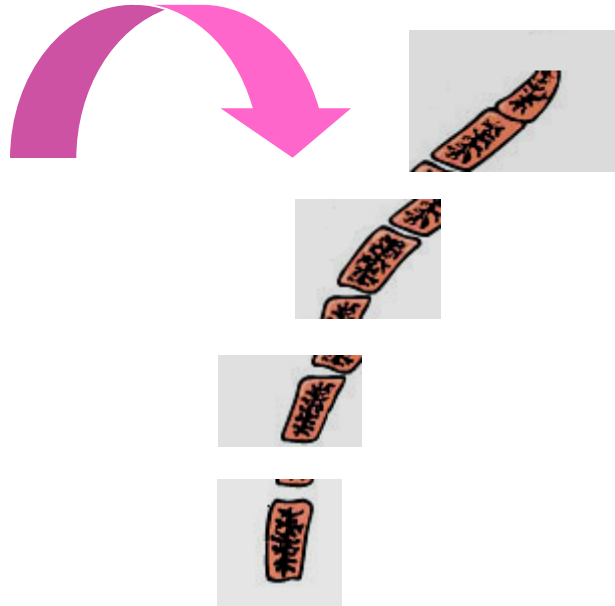
- **Scolex**
 - “Head” of the organism
 - It is the organ of attachment to the intestinal mucosa of the definitive host, human or animal.
- **Neck**
 - Area where new segments are created
- **Strobila**
 - is composed of a chain of proglottids or segments.
 - Series of reproductive organs in various stages.
 -

Strobila = Proglottids = body segments

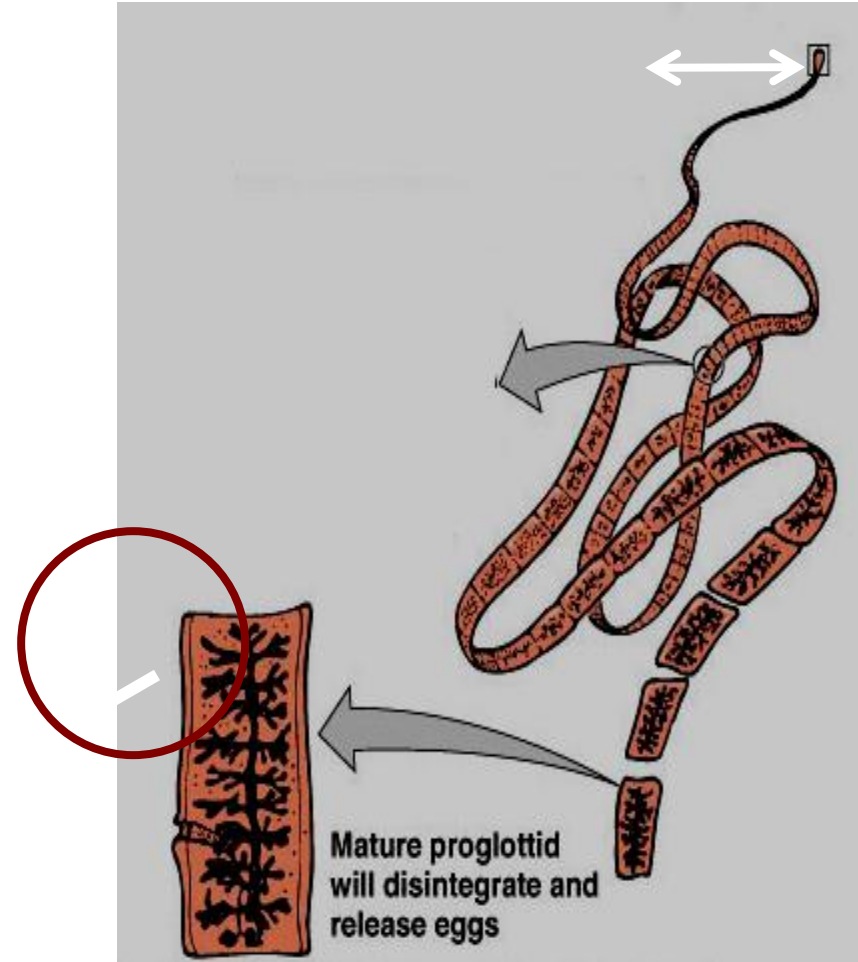


- **Strobilation:** asexual process of forming segments.
- New Strobila are continuously formed in the neck just below the **(A) Scolex**
- Along the length of the worm the Strobila increase in size and maturity, developing from **(B) premature to**
- **(C) mature** carrying fully functional and active sexual organs, to the
- **(D) “gravid” stage** in which essentially the proglottid is filled with the uterus and eggs.

New proglottids arise in the anterior end by a process called strobilation. Segments move posteriorly and mature.

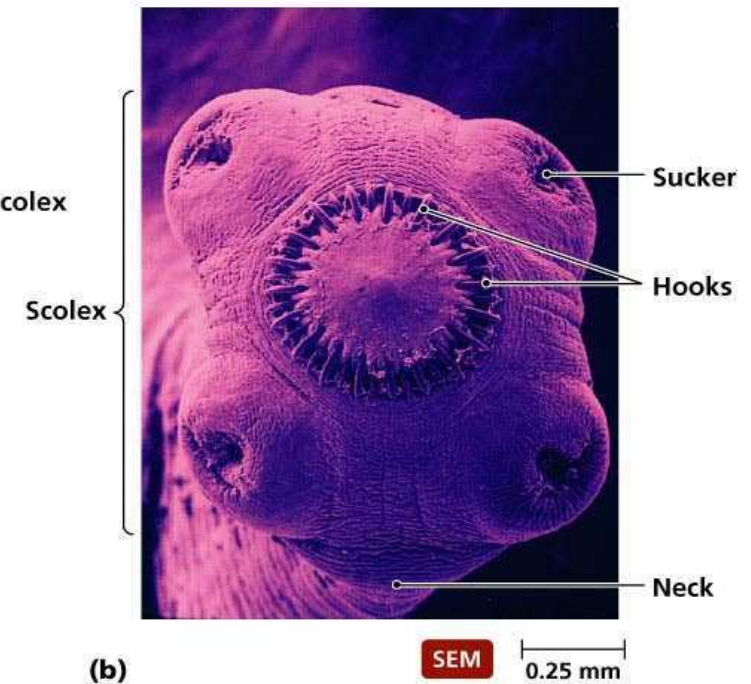
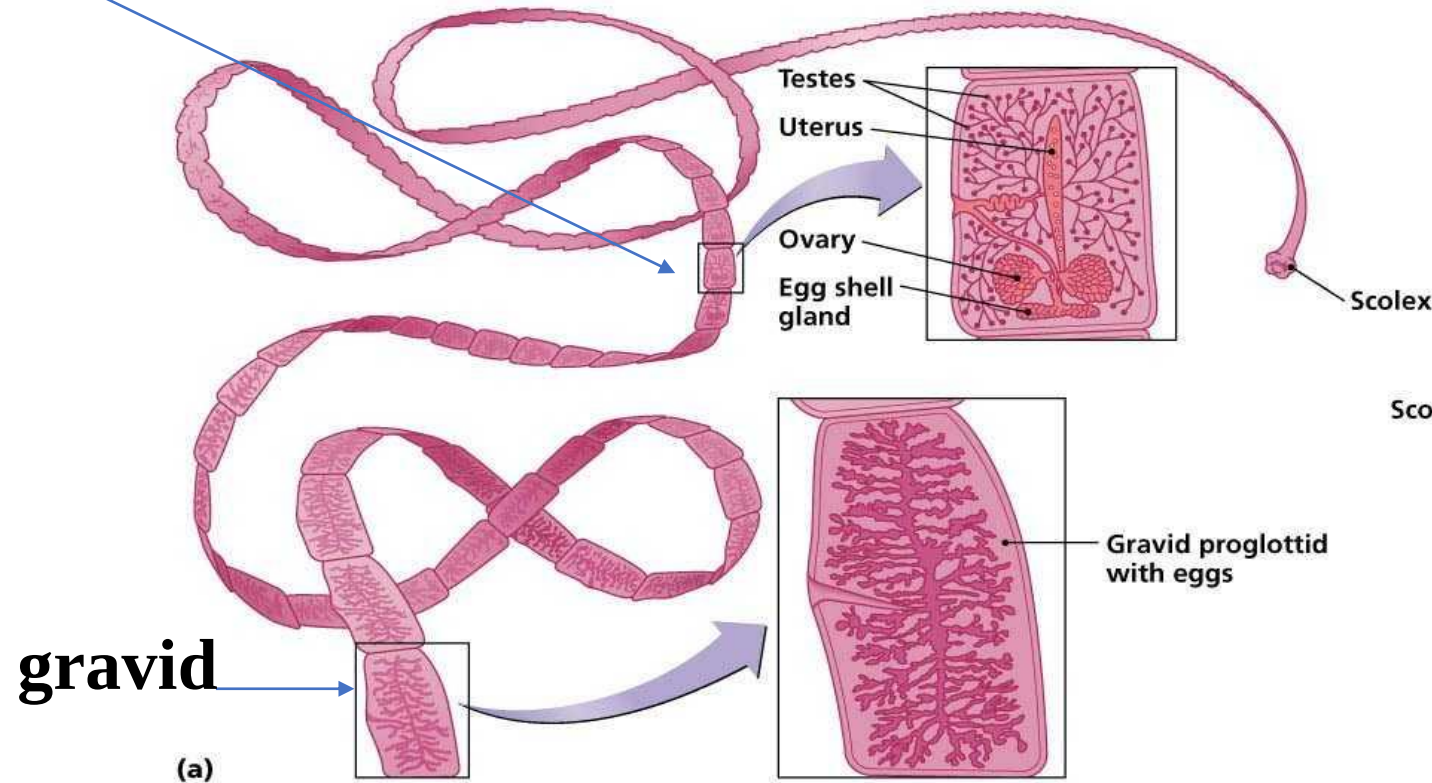


Egg



- The reproductive organs in immature proglottids are visible but non-functional while those in mature proglottids are fully functional, while the gravid proglottids is egg-filled

Mature segment



Medically important cestodes:

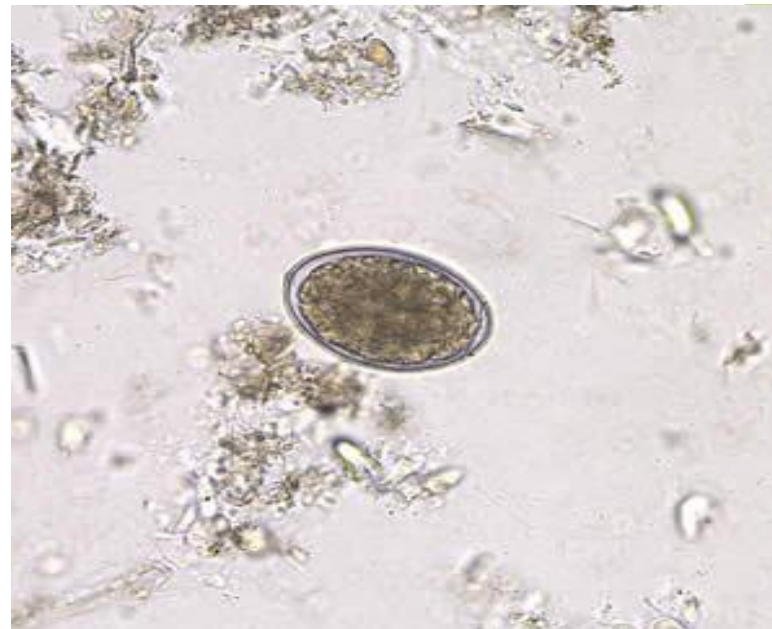
- *Taenia solium* (pork tapeworm).
- *Taenia saginata* (beef tapeworm)
- *Diphyllobothrium latum* (fish tapeworm)
- *Echinococcus granulosus* (hydatid cyst)
- *Echinococcus multilocularis* (alveolar hydatid cyst).
- *Hymenolepis nana* (dwarf tapeworm)

Eggs

- The eggs of Cyclophyllidea and Pseudophyllidea are different from each other.
- The embryo inside the egg is called the **oncosphere** (meaning 'hooked ball') because it is spherical and has hooklets.
- Oncospheres of human tapeworms typically have 3 pairs of hooklets and so, are called **hexacanth (meaning 6-hooked) embryos**.

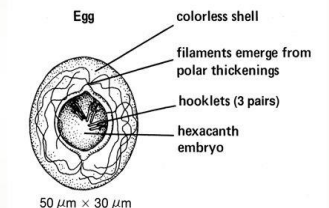


Taenia species



Diphyllobothrium latum

H. Nana egg - Diagnostic Stage



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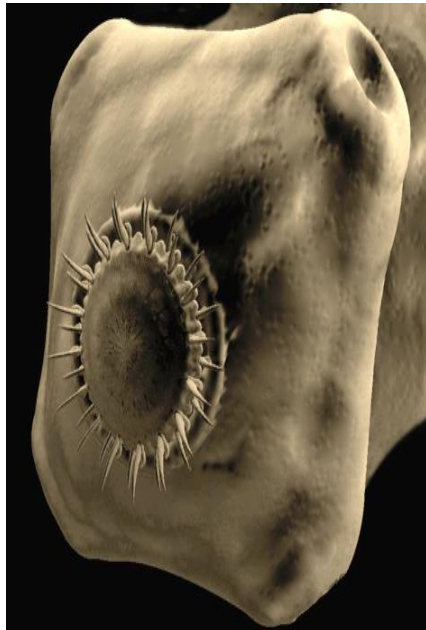
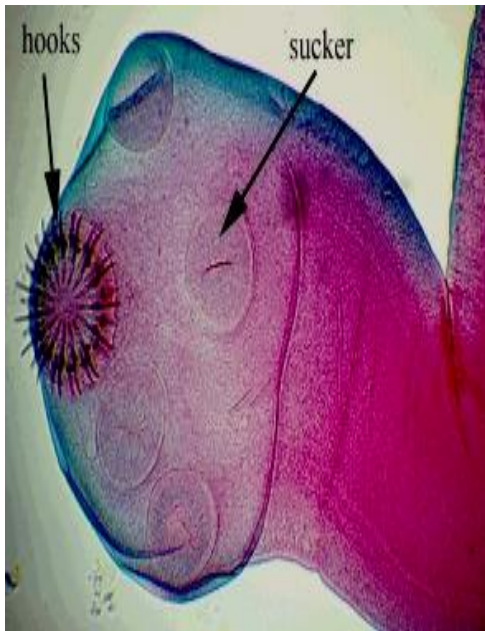
Life Cycle

- Cestodes complete their life cycle in 2 different hosts (Final host and Intermediate host).
- Exceptions are:
- Hymenolepis that requires only **1 host** (Man)
- Diphyllobothrium that requires **3 hosts**, definitive host: **man**; first intermediate host: cyclops; and second intermediate host: fish.
- **Humans are the definitive host** for most tapeworms, which cause human infection.
- An exception is the dog tapeworm (Echinococcus granulosus) for which **dog is the definitive host** and man is the intermediate host.



***Taenia saginata* (Beef tapeworm)**

- **Definitive Host:** Humans
- **Intermediate Host:** Cattle
- **Transmission:** Ingestion of larval form in undercooked beef.
- **Habitat :** Small intestine



***Taenia solium* (Pork tapeworm)**

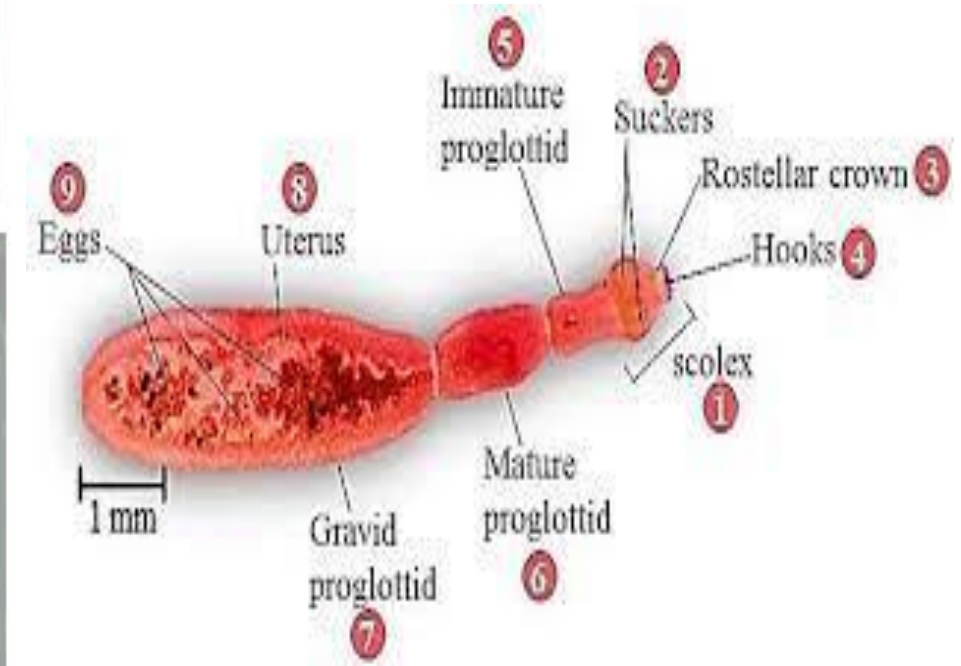
- **Definitive Host:** Humans
- **Intermediate Host:** Pigs
- **Transmission:** Ingestion of larval form in undercooked pork.
- **Habitat :** Small intestine

Adult of *Echinococcus granulosus*

Hydatid Tapeworm

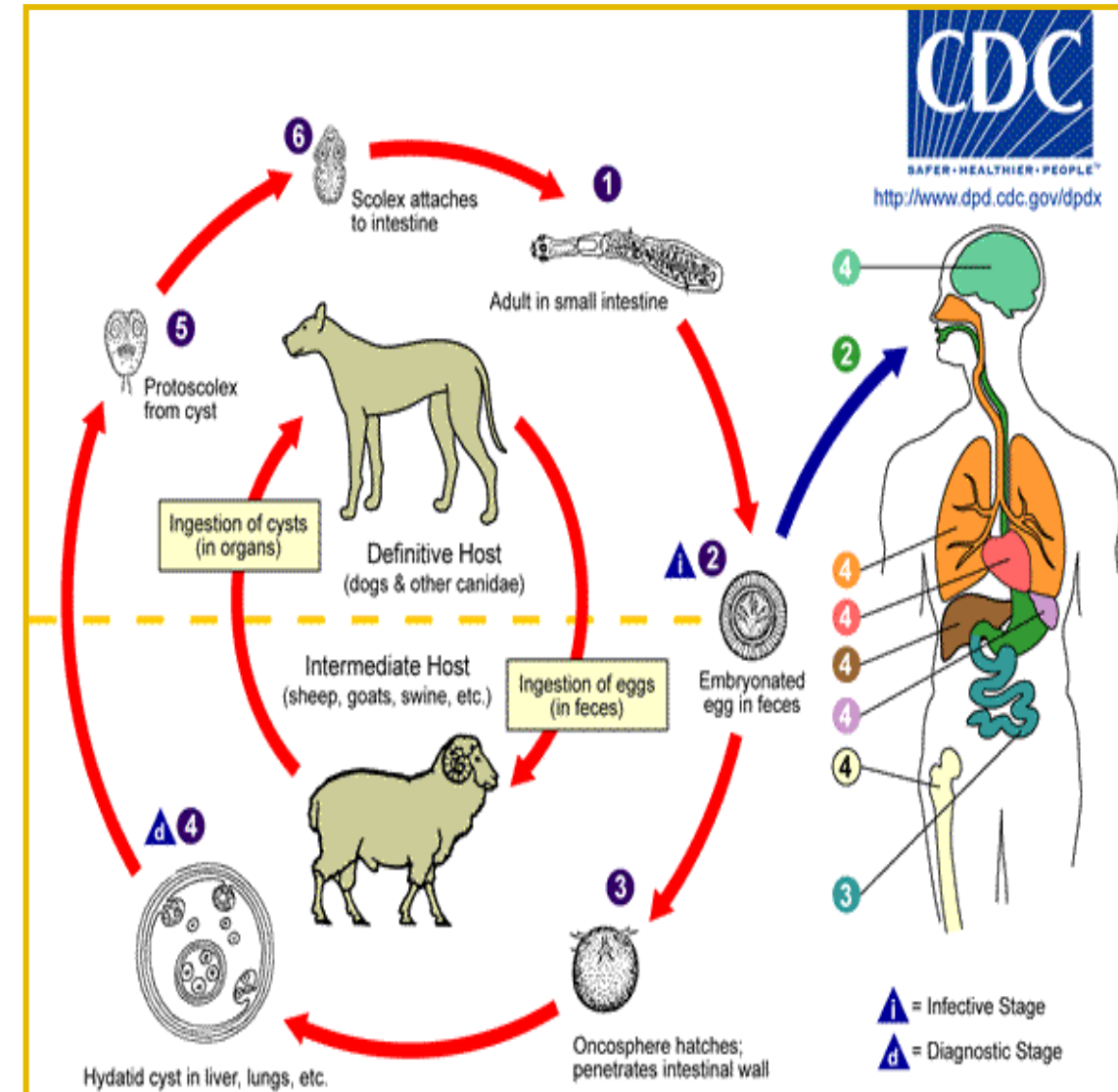
- Scolex, neck, 3 segments
- Egg , hexacanth oncosphere

- **Largest larval stage of all tapeworms:**
Hydatid cyst

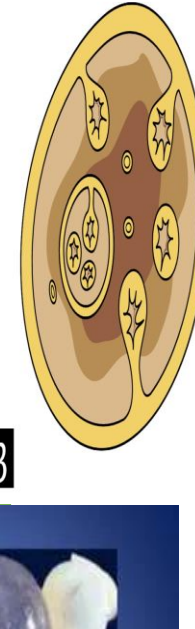
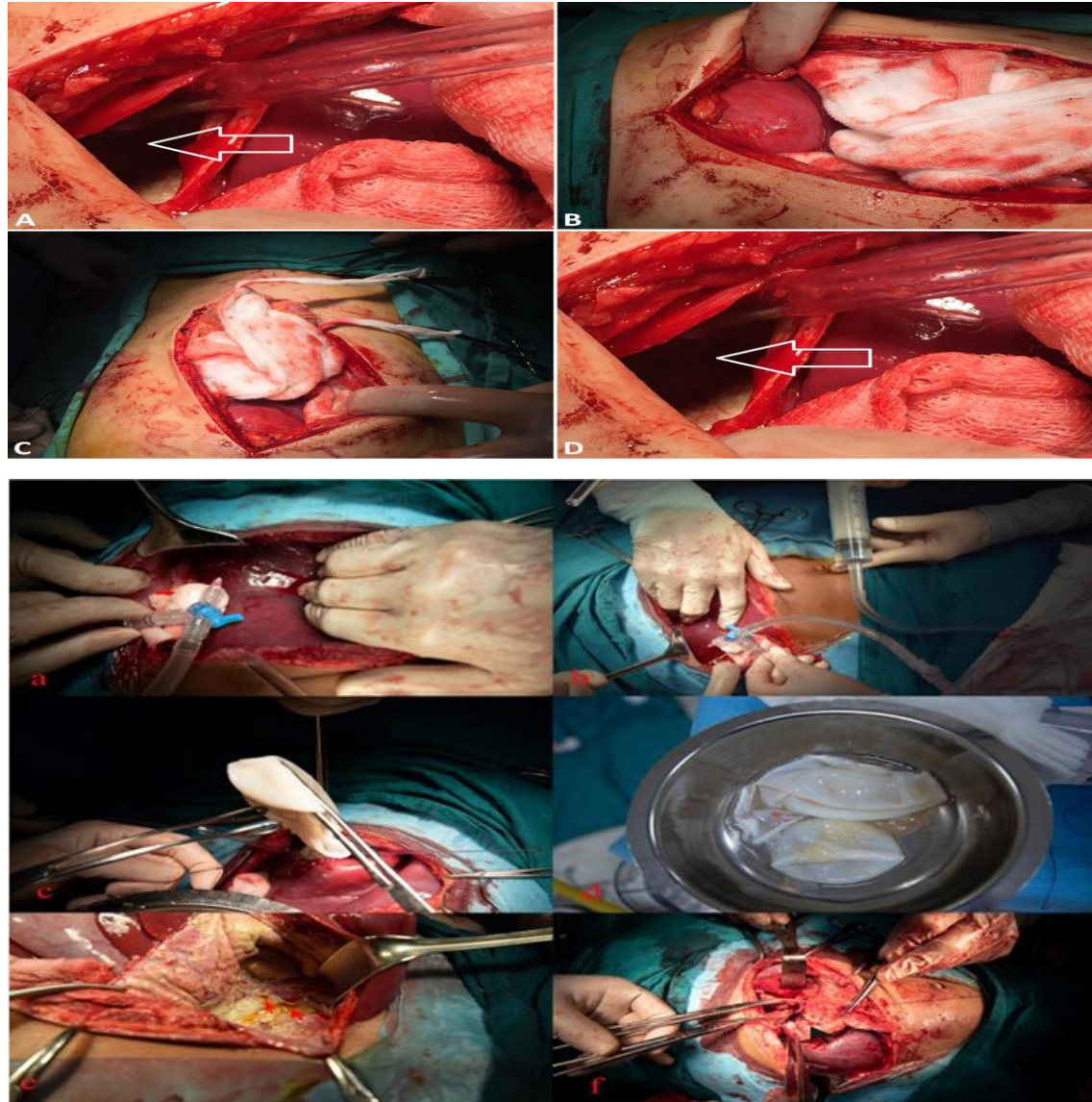


- The adult worm lives for 6–30 months.

- *Echinococcus granulosus* completes its life cycle in 2 hosts:
 - **Definitive host** (dogs) and other canine carnivora wolf, fox...
 - **Intermediate host** (humans) and herbivores animals, sheep, goat, cattle and horse).
 - The larva of *E. granulosus* causes (**Cystic hydatid disease**).

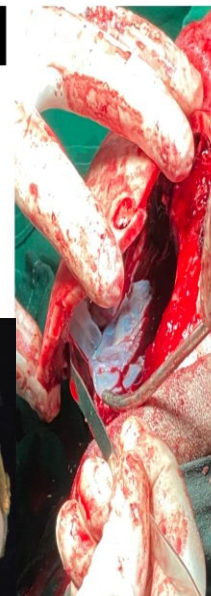
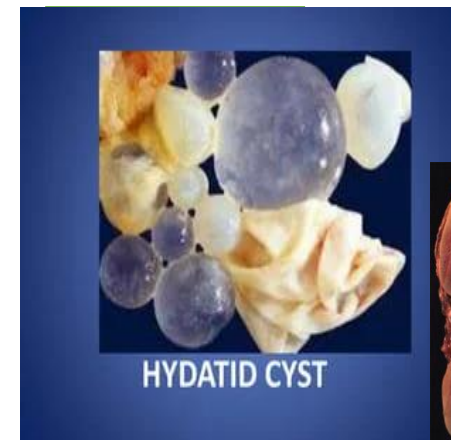
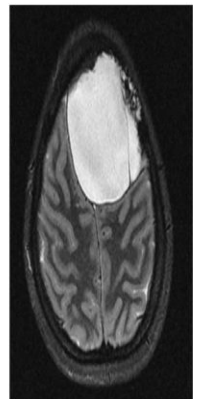
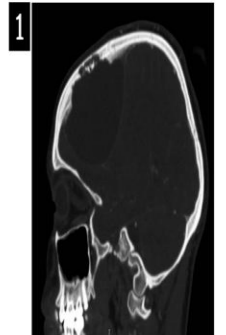


Hydatid cysts in liver



Intracranial-extradural hydatid cyst

The incidence of a
 intracranial localization at
 the level of the skull is below
 0.08%

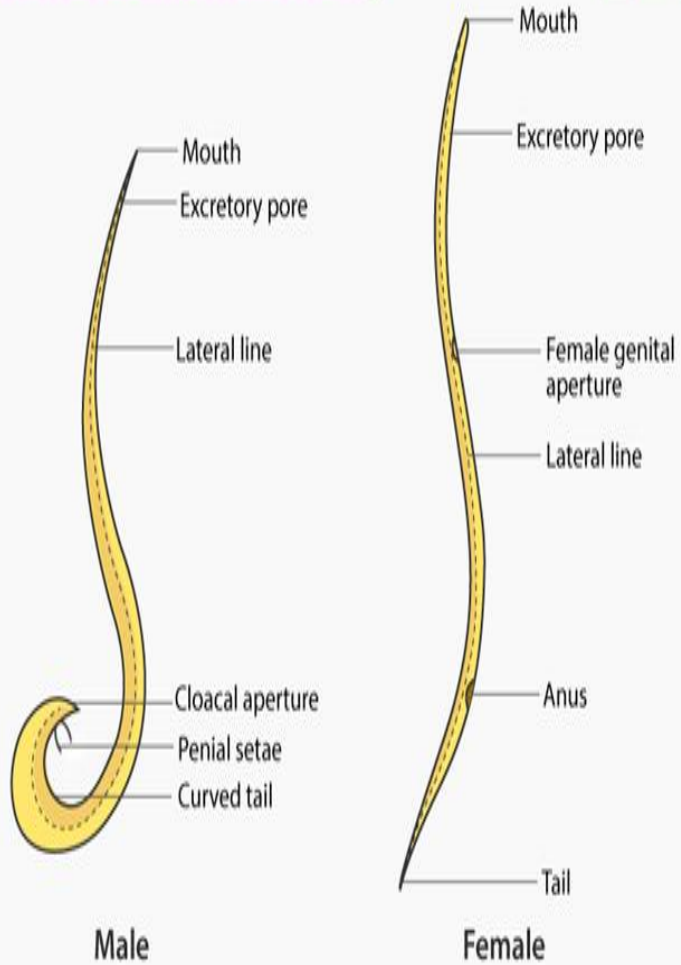


General Characteristics of Nematodes (roundworms)

- Nematodes are elongated, cylindrical, unsegmented worms with tapering ends.
- The name ‘nematode’ means ‘**thread-like**’, from ‘nema’ meaning ‘thread’.
- The adults vary greatly in size, from about a **millimeter** (*Strongyloides stercoralis*) to a **meter** (*Dracunculus medinensis*) in length.
- Male is generally **smaller** than female and its posterior end is curved or coiled ventrally.
- The posterior end of the male is curved ventrally and with two **spicules** (*Ascaris lumbricoides*) or with **bursa** (*Ancylostoma duodenale*) which is used to hold the female during the copulation.

ASCARIS LUMBRICOIDIS

BYJU'S
The Learning App

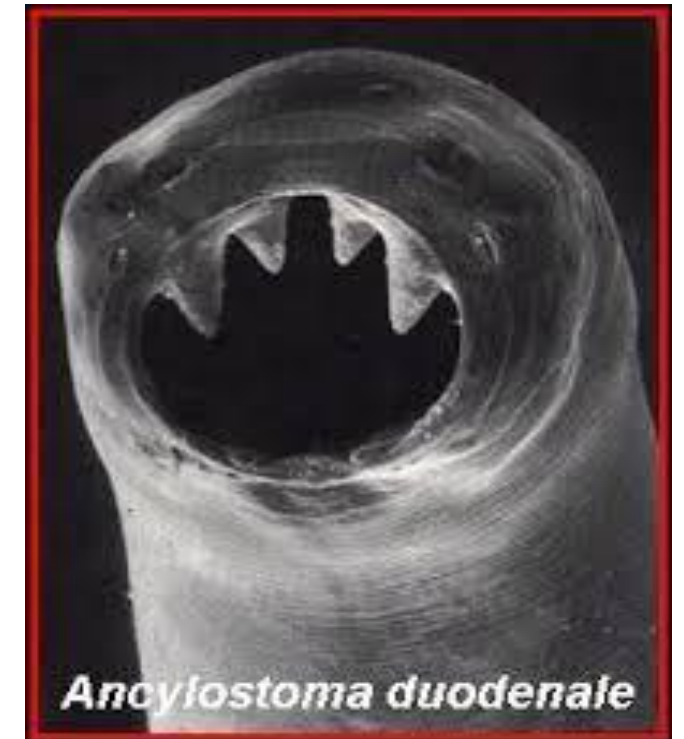
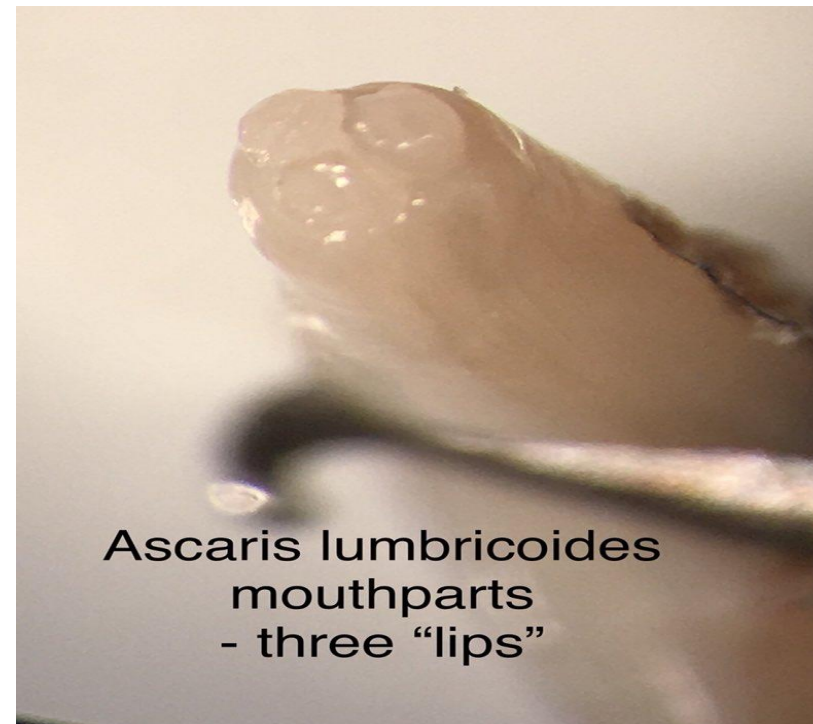
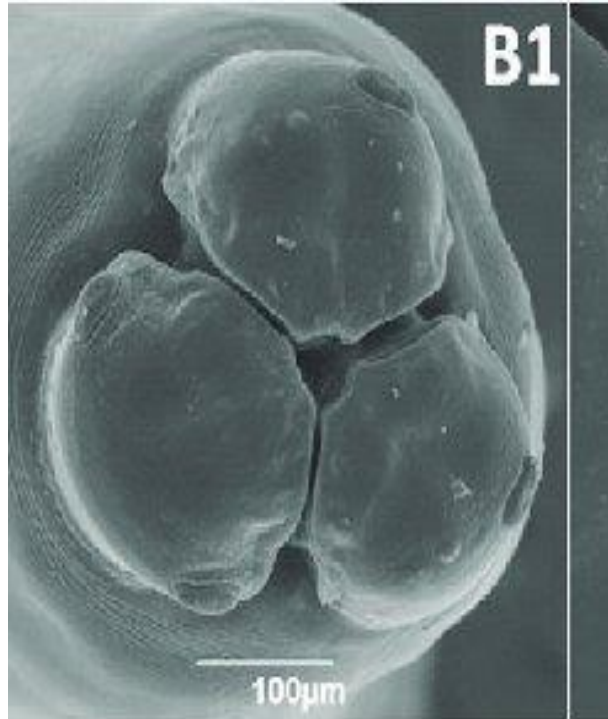


Adult worms of *Ascaris*
Lumbricoides

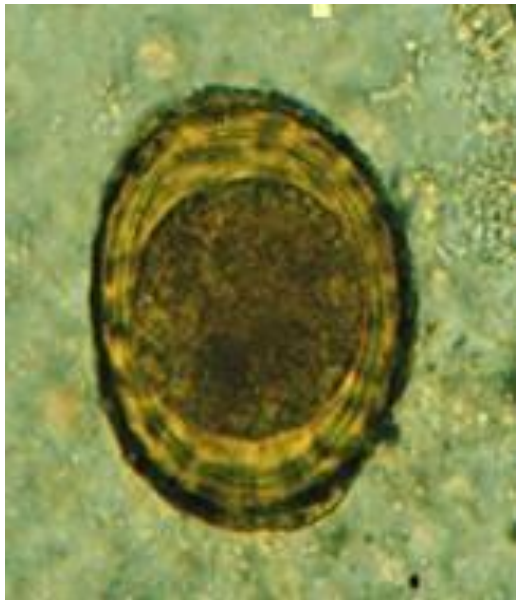
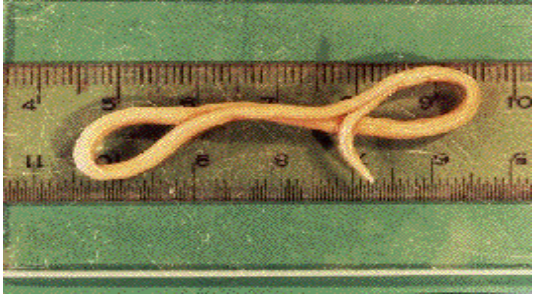


Bursa (Ancylostoma duodenale)

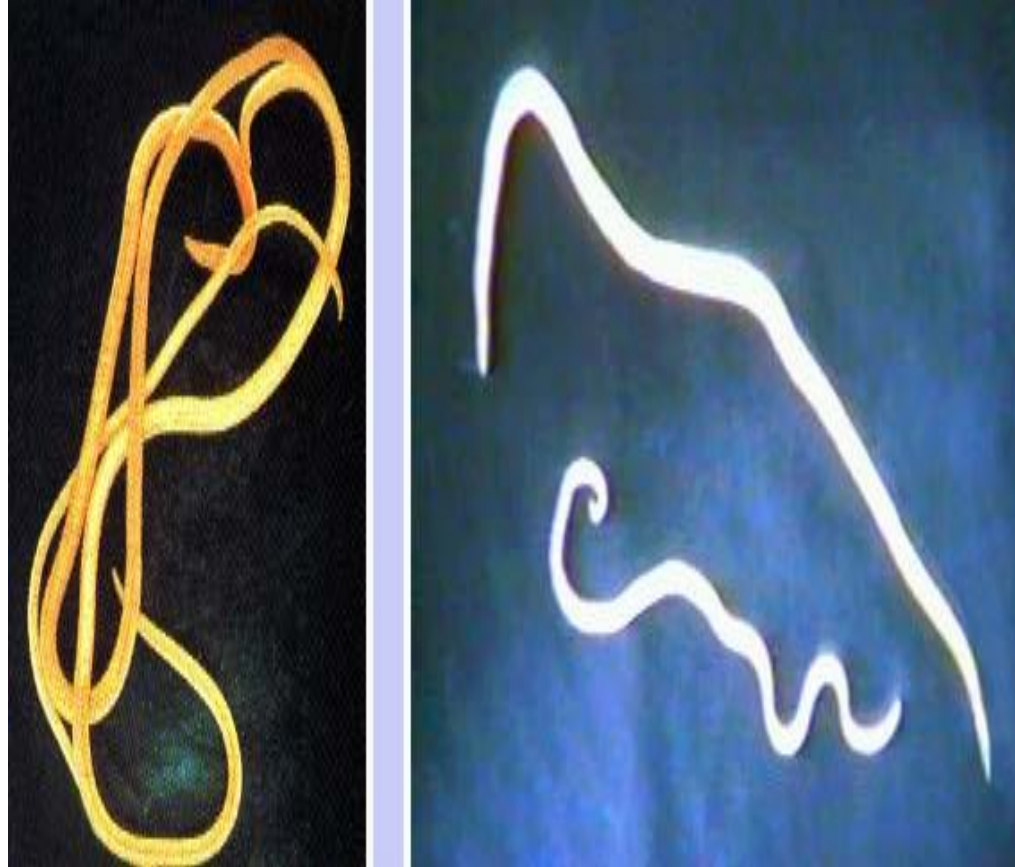
- The adult anterior end called buccal cavity may have teeth or lips. Buccal cavity armed with two ventral pairs of teeth (*Ancylostoma duodenale*) or lips (*Ascaris lumbricoides*). These are used for attachment.



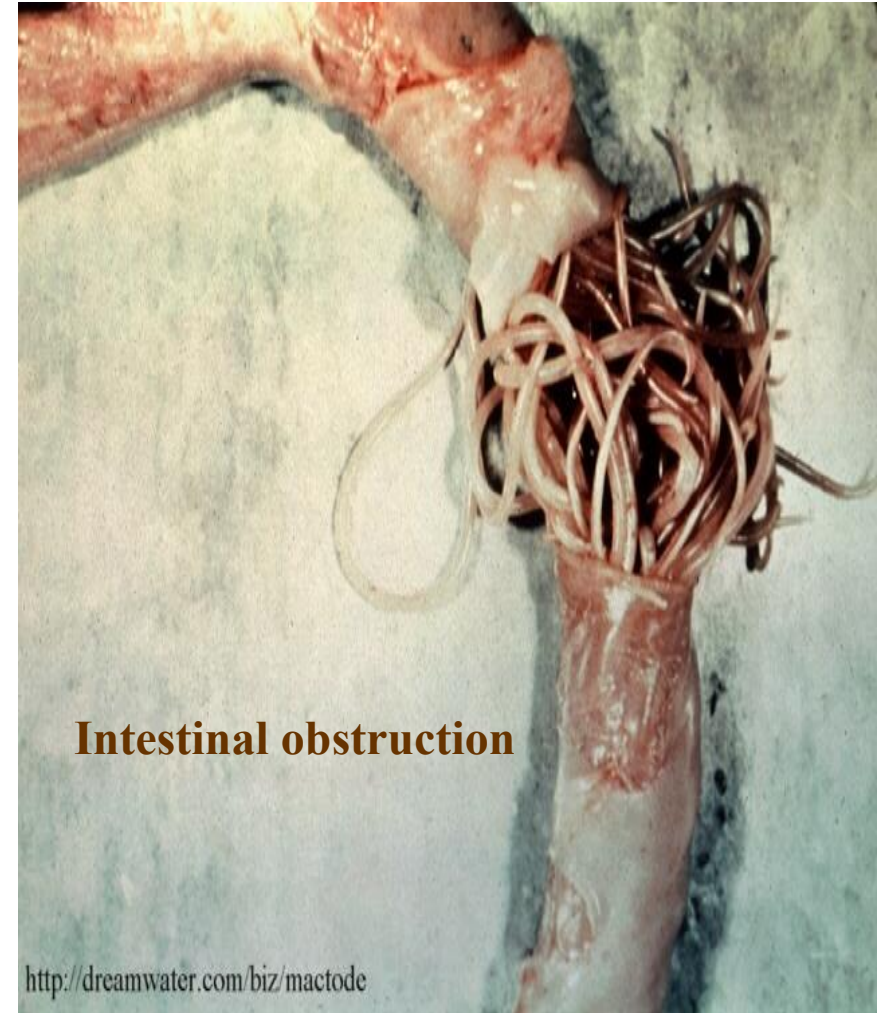
- Their body is covered with **a tough outer cuticle**, which may be smooth, striated, bossed, or spiny.
- The body cavity is a pseudocoel, in which all the viscera are suspended.
- They possess a relatively **well-developed complete alimentary canal, with an anus**.
- Nematodes have simple excretory and nervous systems
- Nematodes **are diecious** (sexes are separate).
- The reproductive system is **complete**.
- Female nematodes may produce eggs (oviparous) or larvae (viviparous). Some lay eggs containing larvae, which immediately hatch out (ovoviviparous)



Fertilized *Ascaris* Egg



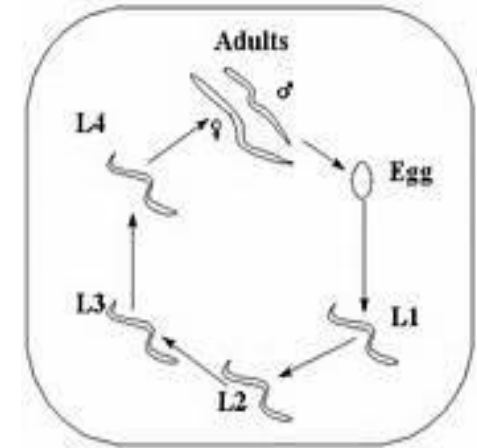
Adult worm of *Ascaris lumbricoides*



Intestinal obstruction

The life cycle

- The life cycle of nematodes consists typically of **4 larval stages and the adult form**.
- The cuticle is shed while passing from one stage to the other.
- Man is the optimum host for nematodes.
- They pass their life cycle in **one host**, except filarial nematodes and *Dracunculus medinensis*, which complete their Life cycle in **two hosts**.
- Arthropods may serve as second hosts or vectors.



Modes of Infection:

By ingestion of:

- 1- **Eggs**: *Ascaris*, *Enterobius*. *Trichuris*
 - 2- **Larvae within intermediate host**: *Dracunculus*
 - 3- **Encysted larvae in muscle**: *Trichinella*
- **By penetration of skin**: *Ancylostoma*, *Necator*, *Strongyloides*
 - **By blood-sucking insects**: *Filariae*
 - **By inhalation of dust containing eggs**: *Ascaris*, *Enterobius*.

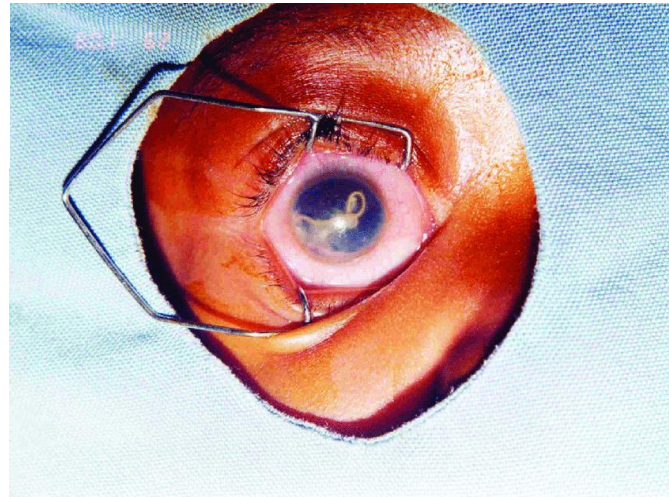
- In most cases, an infection of a small number of worms in a relatively healthy individual may remain asymptomatic or cause minimal discomfort.
- However, patients who have a heavy worm burden, particularly if combined with other health problems, are probably more likely to experience severe symptoms and/or complications.



Rectal prolaps (*Trichuris trichiura*)



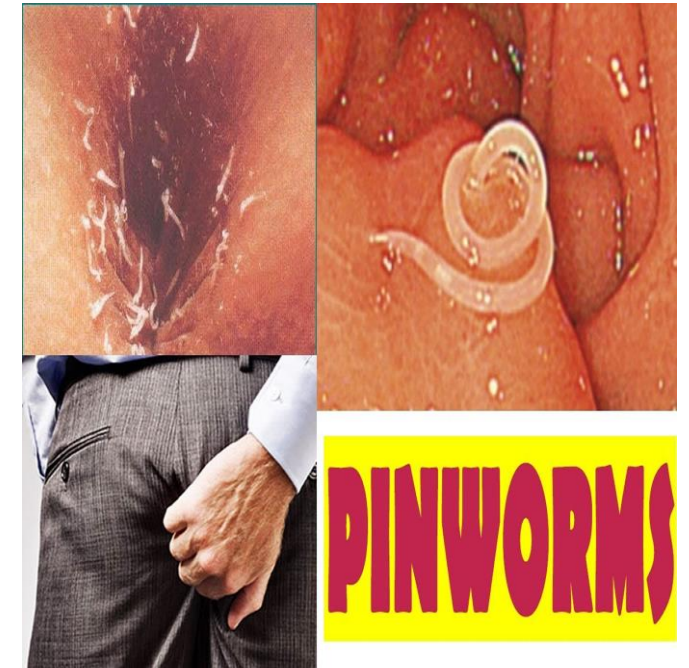
Elephantiasis



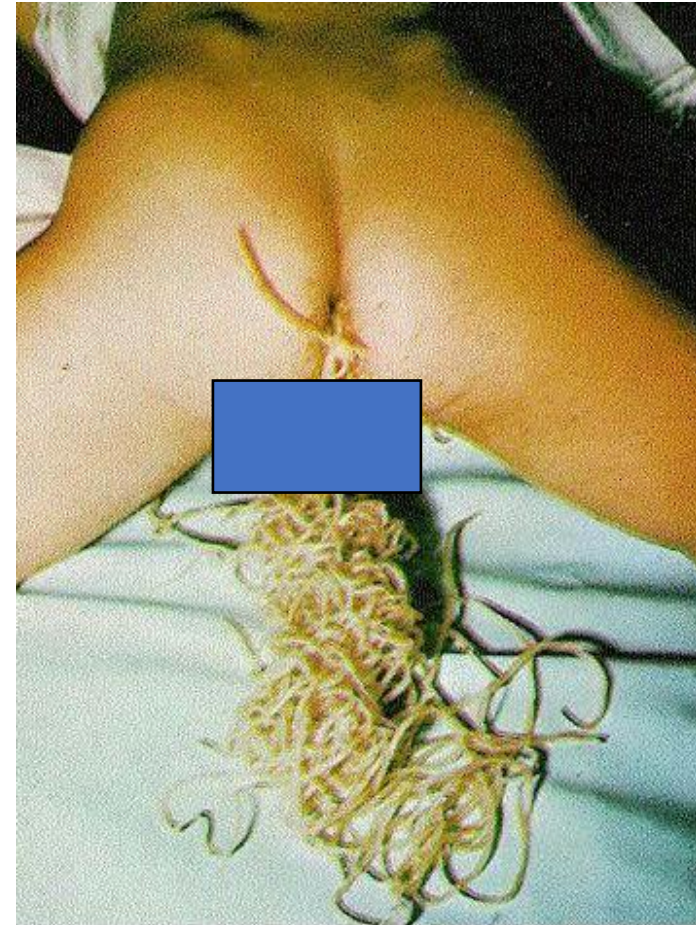
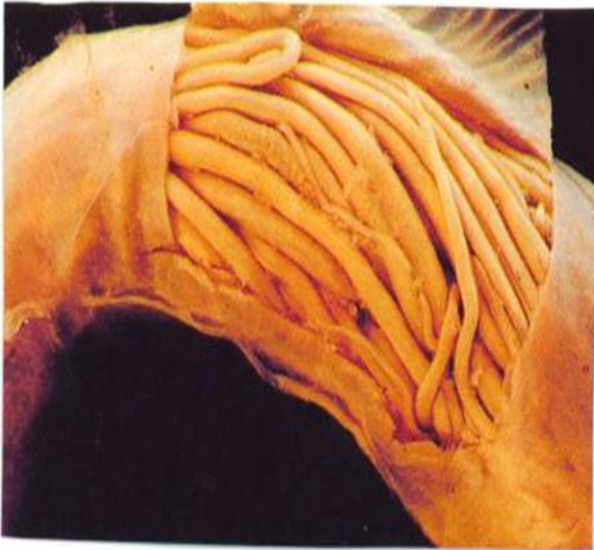
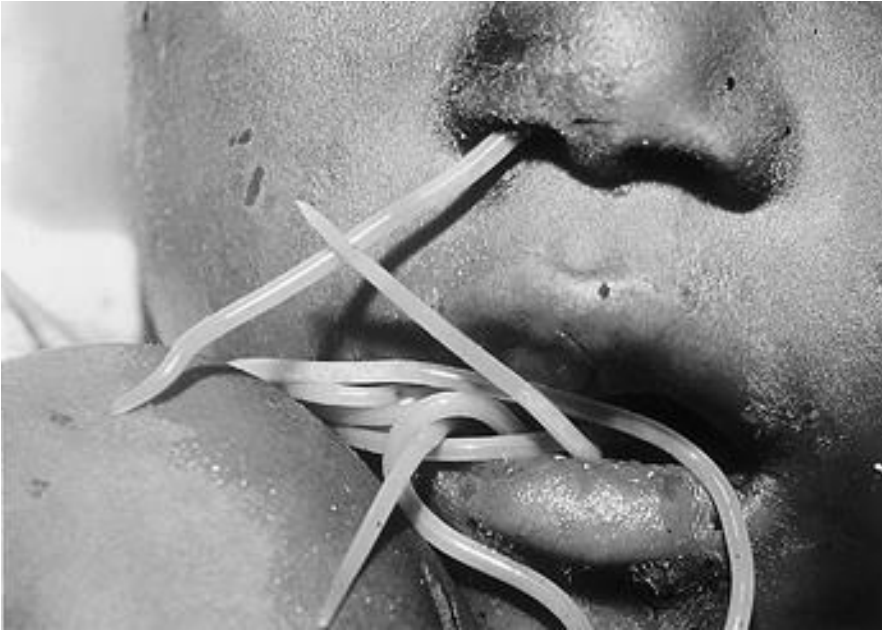
Loa loa conjunctiva



Dracunculus medinensis



Ascaris



Massive *Ascaris* infection in child. A large bolus of roundworms expelled following anthelmintic treatment.

Classification of Nematodes on the Basis of the Habitat of Adult Worms

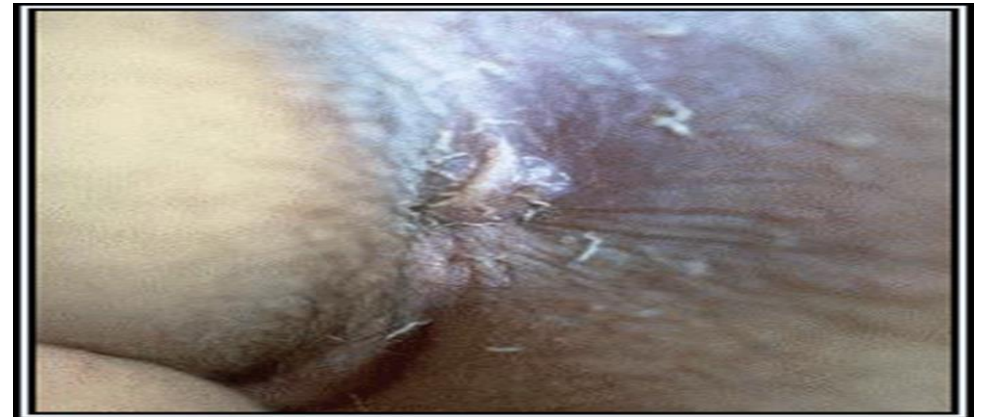
Small Intestine

- *Ascaris lumbricoides* (Common round worm)
- *Ancylostoma duodenale* (Old world Hook worm)
- *Necator americanus* (American or New World Hook worm)
- *Strongyloides stercoralis*
- *Trichinella spiralis*



Large intestine

- *Trichuris trichiura* (Whip worm)
- *Enterobius vermicularis* (Thread or pin worm)



Lymphatics

- *Wuchereria bancrofti* (Elephantiasis)
- *Brugia malayi*
- *Brugia timori*



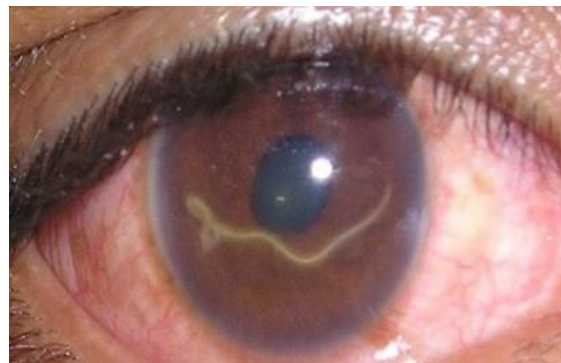
Skin/subcutaneous tissue

- *Onchocerca volvulus* (River blindness)
- *Dracunculus medinensis* (Guinea worm)



Conjunctiva

- *Loa loa* (Eye worm)



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